

Sahaay

सहाय · assistance

Your PC, *hands-free* and *eyes-free*.

An offline accessibility copilot for Snapdragon-powered HP PCs. Six AI models, all on the Hexagon NPU. Nothing leaves the laptop.



TWO PEOPLE

A laptop is only a laptop if you can *move a mouse* and *see a screen*.



Arjun, 22

Engineering student · cerebral palsy

Cannot hold a trackpad steady. Every assignment, portal and chat needs **someone else's hands**.



Meena, 58

Shopkeeper · low vision, diabetic retinopathy

Can hear a screen reader list forty buttons. **Cannot see the photo her daughter sent**, or the chart in her bank app.

Composite personas drawn from published accessibility research; illustrations, not real people.

THE PROBLEM

Assistive technology is priced like a hospital device and built for the cloud.

—

2.68 crore

Indians live with a disability. Roughly one in five with movement, one in five with vision.

Census of India 2011

—

₹1.5 lakh+

for a commercial eye-gaze system, plus a dongle, a licence and often an internet connection.

Typical market pricing, 2026

—

0

cloud assistants that can watch your face all day and read your bank statement without sending both to a server.

Privacy, latency and connectivity

Built-in accessibility reads buttons. It cannot describe a photo, a chart, or the room. *And it still needs your hands.*

THE INSIGHT

The NPU inside a ₹60,000 laptop *is* the assistive device.

A Snapdragon Hexagon NPU runs face tracking, speech, a language model and a vision model at the same time, at a few watts, with nothing leaving the machine. HP ships it in the OmniBook that students already buy.

₹1.5 L+

eye-gaze hardware, licence,
cloud account

₹0

Sahaay: a download for the
laptop you already own



One app. Two modes. *One brain on the NPU.*

HANDS-FREE

Your head is the mouse. Your voice is the keyboard.

For Arjun: motor impairment, injury, RSI

- **Nudge your head**, the cursor moves. Hold still, it stays. Calibrates itself.
- **Blink** to click, **hold still** to click, **open your mouth** to drag.
- **Dictate** into any app, in English or Hindi.
- **Say it:** *"open my downloads", "write a thank-you note to my teacher and save it."*
- **Or press a switch.** Any Arduino becomes a button interface.

NARRATOR

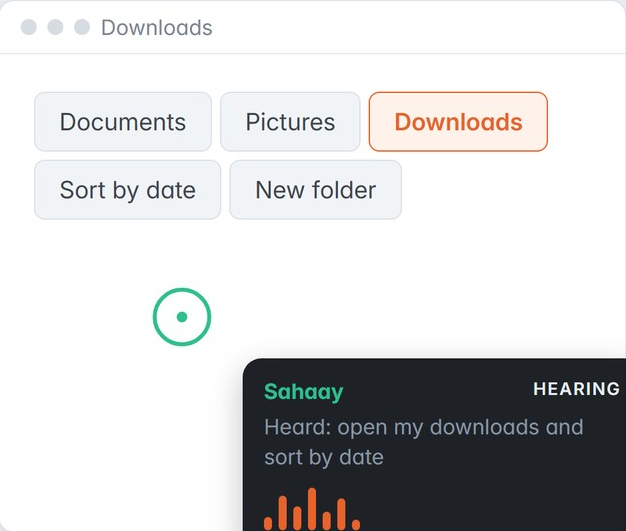
Ask the screen what it says. Ask the camera what it sees.

For Meena: low vision, blindness

- *"What's on my screen?"* Two spoken sentences: **which app, what content, what matters.**
- *"Read this."* The paragraph under focus, in her language.
- *"Describe the picture."* A vision model explains **photos, charts, canvases.**
- *"What do you see?"* The webcam view, labels and signs, **read aloud.**
- Every action is **confirmed by voice**, so she always knows what just happened.

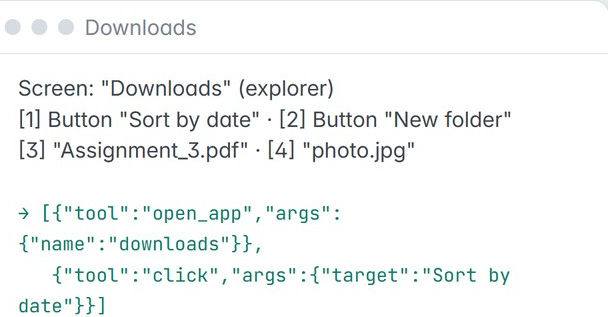
Say it. Sahaay *sees the screen*, plans, and does it. Then tells you.

1



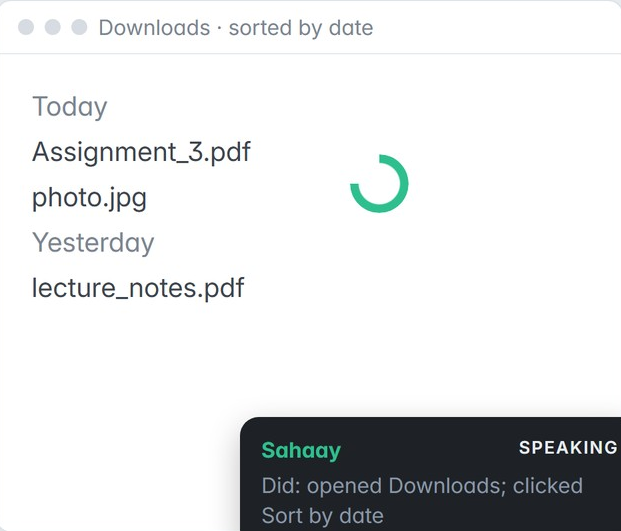
Sahaay HEARING
Heard: open my downloads and sort by date


2



Sahaay THINKING
whisper 79 ms llm 1.4 s face 2.7 ms all on the NPU

3



Sahaay SPEAKING
Did: opened Downloads; clicked Sort by date

ARJUN SAYS

"Open my downloads and sort by date."

ON THE NPU

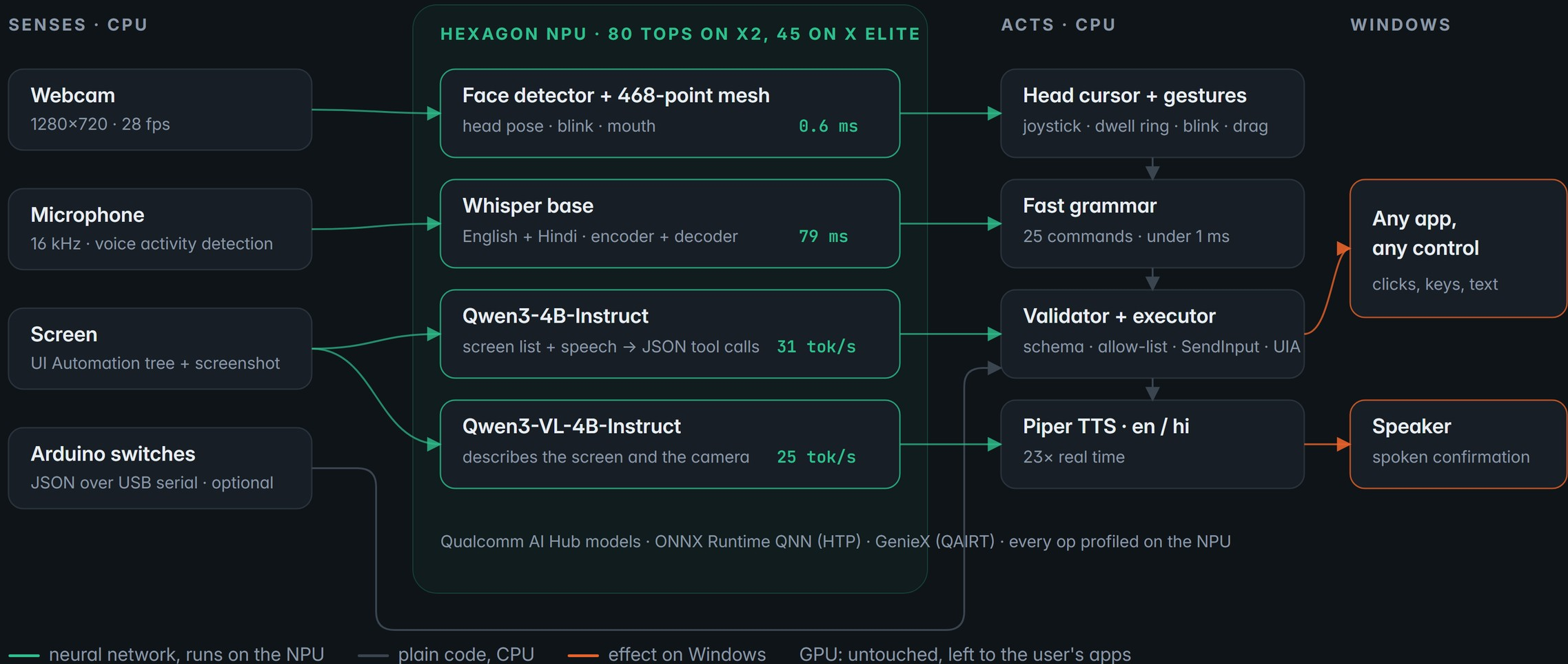
Speech becomes text, the screen becomes a list, the list plus the request becomes two safe tool calls.

SAHAAY SAYS

"Downloads is open, sorted by date. Newest first is Assignment 3."

HOW IT WORKS

Six models on one NPU. *The CPU only moves the mouse.*



PROOF, NOT CLAIMS

Measured on the NPU, *provider asserted* first.

Latency per stage, CPU vs NPU

Median of 30 runs on the development laptop (Snapdragon X2 Elite). Same graph, same inputs.

Face pipeline

detect, crop, 468-pt mesh, pose · live 720p frame



Provider	Latency (ms)
Oryon CPU	25.3
Hexagon NPU via QNN	2.7

Face detector

BlazeFace 256×256



Provider	Latency (ms)
Oryon CPU	2.75
Hexagon NPU via QNN	0.43

Face mesh

468 landmarks 192×192



Provider	Latency (ms)
Oryon CPU	0.87
Hexagon NPU via QNN	0.16

Oryon CPU Hexagon NPU via QNN bars scaled per row

`tools/npu_check.py` asserts the active provider for every model. The usual `providers=[...]` call silently runs on the CPU with today's runtime; Sahaay selects the NPU device explicitly.

Speech and generative models, NPU only

Whisper ships as a precompiled QNN context; the language models run through GenieX on QAI RT.

79 ms

to transcribe a 5.6 s command with Whisper base (21 ms encoder, 2.7 ms per token)

NPU

31 tok/s

Qwen3-4B planning tool calls; 243 ms to first token with the screen in context

NPU

25 tok/s

Qwen3-VL-4B describing the screen or the webcam; 269 ms to first token

NPU

10 s

to load each generative model; face tracking is up in 2 s, so the cursor works almost immediately

NPU

Qualcomm AI Hub, reference devices, every op on the NPU

X2 Elite	face 0.4 ms · mesh 0.2 ms · Whisper enc 21.5 ms · dec 2.4 ms/token
X Elite (HP OmniBook)	face 0.7 ms · mesh 0.3 ms · Whisper enc 45.4 ms · dec 3.7 ms/token

One command to install. *Then it never needs the internet again.*

```
# any Snapdragon X / X2 PC · HP OmniBook, EliteBook · no admin rights
powershell -ExecutionPolicy Bypass -File install.ps1
# Python if missing · venv · models for your chipset, once · espeak-ng · Start Menu
shortcut
```



Private by construction

No account, no cloud, no telemetry. Camera frames, audio and screenshots **stay in RAM** and are discarded. A local log file for support, nothing else written.



Fits the OmniBook a judge owns

Hands-free mode within 16 GB. Both generative models resident on 32 GB (5.4 + 6.1 GB measured), or the 2B vision model on 16 GB. 45 TOPS is enough; 80 TOPS has headroom.



Usable without sight or hands

Global hotkeys, a tray icon, a heads-up panel that shows what was heard and done, **spoken confirmation of every action**, English and Hindi voices, and switch access through any Arduino, including the UNO Q from the AI Lab kit.

Every reason this must run on the laptop is a reason it must run *on the NPU*.

WHAT A DISABLED USER NEEDS	CLOUD ASSISTANT	SAHAAY ON THE HEXAGON NPU
A camera on your face all day, a screen with your bank statement	Uploaded, frame by frame	Never leaves RAM
Cursor latency a person can tolerate	100 ms and up, network dependent	2.7 ms per frame
Hostel Wi-Fi, a hospital ward, a train, a village on 4G	Stops working	Fully offline
A laptop that is your only interface must last the day	CPU or GPU inference drains it	NPU runs the whole stack at a few watts
Cost to a student	Subscription plus per-minute speech APIs	Zero, forever

HP India expects AI PCs to reach half of the country’s PC sales within a few years, and aims the OmniBook 3 and 5 at students.

HP India launch coverage, 2024 to 2026, paraphrased

Qualcomm India’s stated goal is Snapdragon PCs below ₹60,000, with on-device AI that does not depend on data costs or bandwidth.

Qualcomm India interviews on the Snapdragon X roadmap, paraphrased

HOW IT WAS BUILT

Three days, one laptop. *What was easy, what was hard.*

- **Prove the toolchain**

ONNX Runtime QNN on ARM64 Python, AI Hub, GenieX. Found the silent CPU fallback and fixed it in npu.py before writing a line of product.

- **Face, then cursor**

AI Hub export, Qualcomm's MediaPipe pipeline re-done in numpy, head pose and blink metrics; joystick mapping, dwell ring, gestures, calibration.

- **Speech, then brain**

Precompiled Whisper with a KV-cache decode loop; two-tier commands; Qwen3-4B with a strict tool schema and a UI Automation snapshot.

- **Narrator, app, deployment**

Qwen3-VL, Piper in English and Hindi, overlay, tray, hotkeys, logging, installer, model fetcher, benchmark suite, Arduino bridge.

EASY

Qualcomm AI Hub. Every model compiled and profiled on real X2 Elite and X Elite devices in minutes, with per-op compute-unit reports.

GenieX made the language and vision models a pip install and one pull, on the NPU first try.

HARD

Windows ARM64 plumbing. No OpenCV wheel, so the camera goes through DirectShow COM. Piper's espeak bridge is not built for ARM64. QNN context caches must be reused, not regenerated. Two GenieX models cannot load at the same moment. And **the QNN provider must be chosen through the new device API**, or ONNX Runtime quietly runs on the CPU while reporting nothing.

NEXT

Sahaay works today. Here is what *the next quarter* looks like.

- **Gaze on the NPU.** The AI Hub eyegaze model for coarse screen-region selection alongside head pose, for people with very limited neck movement.
- **Piper on the NPU, Hindi and Hinglish commands.** The AI Hub Piper recipe, and a command grammar that understands "neeche scroll karo".
- **Grounding from the vision model.** "Click the blue button" when an app has no accessibility tree: games, canvases, remote desktops.
- **Trials with an assistive-technology centre.** Calibration profiles for tremor and limited range; an MSIX package for the Microsoft Store.

A ₹60,000 laptop that *anyone* can use, because the NPU is already inside it.

Everything in this deck is in the repository: code, benchmarks, installer, model packs and the reviewer tools that prove the NPU is doing the work.

github.com/Vision-jarvis/sahaay

Ruhan Srivastava

IIT Kharagpur · built on a Snapdragon X2 Elite with Qualcomm AI Hub, ONNX Runtime QNN and GenieX